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MyDesign Portfolio

Element A

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

Many schools still rely on paper passes or verbal permission systems when students leave class to go to locations such as the bathroom, nurse, counseling office, or library. These traditional systems make it difficult for teachers and administrators to verify whether students actually arrive at their intended destinations or return within a reasonable amount of time. As a result, schools face issues involving student accountability, lost instructional time, hallway congestion, and safety concerns. The purpose of this project is to develop a MATLAB and Arduino-based RFID student pass tracking system that allows teachers to monitor student movement and determine whether students arrive at approved locations on time.

Many schools still rely on paper passes or verbal permission systems when students leave class for locations such as the bathroom, nurse, or counseling office. These traditional systems make it difficult for teachers to confirm whether students actually arrive at their intended destination or return within an appropriate amount of time. As a result, schools face problems involving lost instructional time, hallway congestion, and reduced student accountability. Recent electronic hall pass systems have been introduced in some schools to improve hallway monitoring and increase safety through real-time tracking features. This issue primarily affects middle and high schools, where students frequently move throughout the building during class periods and teachers must manage multiple pass requests at once.

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This problem is connected to larger issues involving school safety, classroom engagement, and student attendance. When students spend excessive time outside the classroom, they miss instruction and may fall behind academically. Hallway congestion and unsupervised gatherings can also contribute to vandalism, unsafe behavior, and disciplinary incidents. Schools additionally face challenges balancing student accountability with privacy concerns when implementing digital monitoring systems.

A successful prototype could help improve school efficiency by providing teachers with a more organized and automated way to manage passes. Secondary stakeholders such as administrators, security staff, parents, and counselors may benefit from improved student accountability and safer school environments. Additionally, replacing paper passes with digital systems may reduce material waste and simplify data collection related to student movement patterns.

Stakeholder Concerns and Connections (A3, A4)

1. Students (Primary Stakeholder)

Students are directly affected because they use the hall pass system daily. They are concerned about fairness, privacy, ease of use, and avoiding unnecessary restrictions while still maintaining access to important locations like bathrooms or the nurse.

2. Teachers (Primary Stakeholder)

Teachers need a reliable way to monitor students who leave class. They are concerned about lost instructional time, student accountability, and managing multiple pass requests efficiently during lessons.



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3. School Administrators (Primary Stakeholder)

Administrators are responsible for maintaining school safety and reducing hallway disruptions. They need systems that improve accountability while also being practical and cost-effective for schools to implement.

4. Security Staff and Hall Monitors (Secondary Stakeholder)

Security staff benefit from having clearer information about which students should be in hallways during class periods. Improved tracking can help reduce unauthorized movement and improve safety monitoring.

5. Parents and Guardians (Secondary Stakeholder)

Parents are indirectly affected because they expect schools to maintain safe and organized environments for students. Improved tracking systems may increase parent confidence in school safety procedures.

Problem Importance (A6)

The problem of ineffective student pass tracking is important because it directly impacts classroom learning, student safety, and school accountability. Research from the National Center for Education Statistics found that over 93% of public schools reported using some form of security or student monitoring measure during the 2021–2022 school year, showing that schools are increasingly focused on improving student supervision and accountability (National Center for Education Statistics, 2022). Schools that have implemented digital hall pass systems have reported reductions in hallway congestion, unauthorized movement, and excessive time spent outside the classroom, all of which negatively affect instructional time and student performance (Hedes, 2026). In addition, concerns about student safety and monitoring have grown due to issues such as



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vandalism, vaping, and unsupervised hallway gatherings during class periods (Knutsson, 2026). Developing a MATLAB and Arduino-based RFID tracking system provides a practical engineering solution that can improve accountability while also demonstrating how technology can address real-world problems in education.

Reference List:

["Digital Hall Passes for Schools | Student Accountability Software." SmartPass](#)

[. Accessed 11 May 2026.](#)

["Fast Facts: School Safety and Security Measures." National Center for Education Statistics](#)

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[Hedes, Jarrad. "JTASD Approves Electronic Hall Pass System." Times News Online](#)

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[Knutsson, Kurt. "NYC Schools Track Bathroom Time with Digital Hall Passes." CyberGuy](#)

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["Cloud-Based K-12 Student Behavior Tracking System." IdentiSys](#)

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